The Architecture of Mars Global Localization for the Perseverance Rover: Onboard Map Matching and Hardware Repurposing

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Technische Hochschule Würzburg-Schweinfurt (THWS)

Authors

Vihaan Yogish — Student ID: 10007352
Ahillan Mohankumar — Student ID: 10007333
Vineet Mukund Khare — Student ID: 10005851
Nischal Shivanand Indi — Student ID: 10005878
Mokshitha Karnati — Student ID: 10005821

Supervisor: Prof. Dr. Marian Daun

Abstract

Mars Global Localization enables the Perseverance rover to autonomously determine its position by matching panoramic camera images with orbital maps, eliminating Earth-based manual navigation. This paper describes a high-speed processor to run onboard image-matching software that compares panoramic rover images with orbital maps, enabling sub-meter autonomous localization and measurably improving the accuracy and operational efficiency of the Mars rover.

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1 Introduction

Mars rover navigation has traditionally relied on incremental odometry combined with periodic Earth-based localization updates. In this approach, the positional estimates accumulate error over time and must be manually corrected by assisted alignment with orbital imagery. Due to the significant communication latency resulting from the interplanetary distance between Earth and Mars and the limited onboard processing capabilities, the workflow introduces delays and constraints on autonomous mobility on the Martian surface.

The aim of this paper is to examine the implementation and impact of Mars Global Localization (MGL) as a solution to these limitations. MGL enables the Perseverance rover to autonomously determine its position by matching panoramic images captured by onboard navigation cameras with high-resolution orbital maps. The objective is to reduce dependence on Earth-based processing, fix the odometry drift, and eliminate communication bottlenecks in routine navigation. By analyzing the hardware adaptation and software methodology behind MGL, this article evaluates how increased onboard autonomy improves position accuracy, operational efficiency, and overall mission capability.

2 The Problem: Odometry Drift and the "Ground-in-the-Loop" Delay

Initial Mars rovers such as Sojourner rover (NASA's Mars Pathfinder mission July 1997) worked based on a Waypoint-Based Navigation system that relied primarily on Two high-resolution color stereo cameras in combination with light-striper sensors to detect hazards [2]. This navigation system was supported by dead reckoning methods for navigation, That is, estimates a moving object's current position and pose based on a known starting point, speed, direction, and elapsed time. It does not rely on external references such as stars, landmarks, or GPS. In the Sojourner rover, dead reckoning was performed by combining Inertial Measurement Units (IMUs - gyro) and wheel odometry (WO - wheel encoders) measurements.

This method is affected by significant errors associated with the slippage of wheels on loose soil that accumulate over time.

2.1 Possible solutions implemented over the years in various previous rovers

To compensate for dead-reckoning errors, Visual Odometry (VO) techniques allows highly accurate pose estimates of moving assets by tracking fiducial points (reference points) of a scene observed by the onboard cameras [2].

VO was first used for planetary applications by the Mars Exploration Rovers (MER) Spirit and Opportunity. The MER-VO algorithm was based on the determination of 3D coordinates of selected keypoints after stereo-matching of left- and right-eye images through correlation methods giving an accuracy of 2 mm [1]. However, due to the limited computing capabilities of the rover this algorithm required 2-3 minutes for each ground step, reducing the speed of the rover to 8% of the maximum speed [2].

Such VO algorithms first became effective in the next Mars rover, Curiosity, where a refined stereo correlation algorithm and an iterative image pyramid scheme were included in the VO algorithm. Iteratively at each level of the image pyramid, the MSL-VO algorithm allows to constrain the search of tracked features in the finer resolution images [2,3]. This algorithm then led to obtain a motion estimate in 47 seconds on average, much faster than before.

The NASA's MER (Mars Exploration Rover Opportunity (MER) Opportunity) and MSL missions therefore became testing grounds for creating a solid technique for accurate localization of planetary rovers through VO algorithms. These algorithms and techniques of using the dead reckoning method with VO algorithms became the foundations for the Navigation of the Perseverance Rover.

2.2 The case of Perseverance rover

Perseverance is much more advanced compared to its previous counterparts using virtually odometry to track geological features from the very beginning, but during the processing of camera images and wheel slippage, interplanetary errors persist [4]. For the perseverance rover these small errors from slippage and terrane inconsistencies build up over time on longer drives by up to more than 100 feet (35 meters) [4]. This leads the rover to believe that it is in hazardous terrain, end its drive prematurely, and wait for instructions from the Earth. Once the rover stopped, it would send a 360-degree panorama to Earth, where experts match the imagery with shots from NASA's Mars Reconnaissance Orbiter (MRO). The team then sends the rover its location and instructions for its next drive. That process can take a day or more [4].

3 The Hardware: Repurposing Ingenuity's Brain

The RAD750 is a radiation-hardened single-board computer manufactured by BAE Systems Electronics, Intelligence and Support. The successor of the RAD6000, the RAD750 is for use in high-radiation environments experienced on board satellites and spacecraft. The RAD750 was released in 2001, with the first units launched into space in 2005.

It has a core clock of 110 to 200 MHz and can process at 266 MIPS or more. The CPU can include an extended L2 cache to improve performance. The CPU can withstand an absorbed radiation dose of 2,000 to 10,000 grays (200,000 to 1,000,000 rads), temperatures between 55 °C and 125 °C, and requires 5 watts of power.

The Perseverance rover is the first rover to be able to drive autonomously at close to its maximum electromechanical speed and its self-driving autonomous navigation system has been used to evaluate 88% of the distance traveled. The greatest distance it has driven without human review is 699.9m over three days which is a planetary rover record. This overall distance is limited by the growth in the uncertainty of the position, shown in Figure 1. [7]

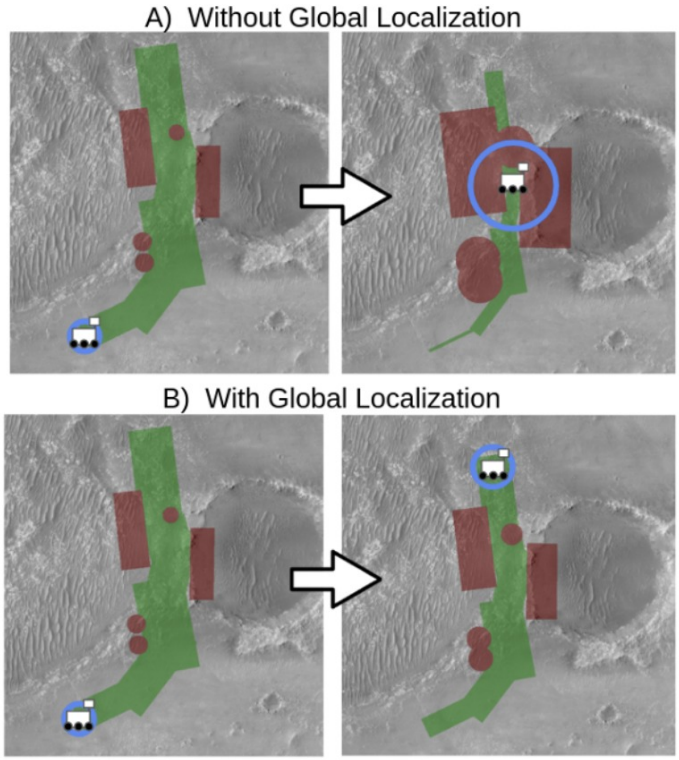


Figure 1: Growth of rover position uncertainty during long autonomous traverses.

3.1 Computational Requirements of Localization

Rover localization is modeled as an image registration problem between the rover images and the orbital map. A model-based approach is taken to make the rover images match the orbital map image as closely as possible, then find a similarity measure that is invariant to the differences [7]. The following subsections detail the stages of the approach, depicted in Figure 2:

- *Absolute Orientation Estimation:* Rover missions can typically estimate absolute orientation with high accuracy, either by estimating rover orientation based on the position of the sun, the stars, or internally through gyrocompassing. By running one of these activities in advance, the registration problem can be simplified to a search over 2D translation.
- *Panoramic Stereo Imaging:* A 360 panorama of stereo pairs around the rover is acquired to maximize the area of imaged terrain. For Perseverance, monochrome red channel images are used, which are spectrally the most similar to the red/near-infrared channel HiRISE map.

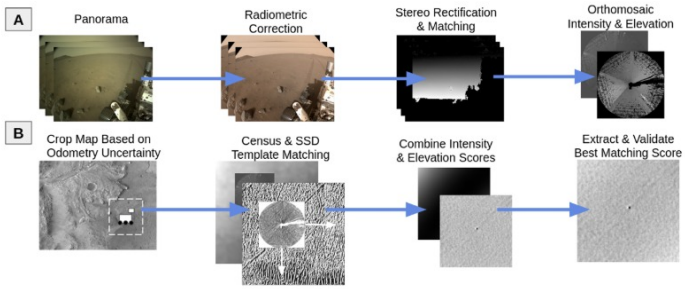


Figure 2: High-level overview of the global localization framework. A) Transforming the rover images to match the orbital map: a 360 rover panorama of stereo images are radiometrically corrected, stereo matched, and orthographically projected into a 2D grid. B) Matching the rover images to the orbital map: the orbital map is cropped based on rover odometry's position uncertainty, the orthomosaics are template matched to the orbital map based on the census and SSD similarity measures, and the best score is extracted and converted into a delta position update.

3.2 Repurposing Ingenuity's Processor

The fast processor on the Ingenuity Helicopter Base is mounted on the Perseverance rover, as an auxiliary computing resource to perform the localisation. The Ingenuity system utilizes a commercial Qualcomm Snapdragon 801 processor, which delivers a higher computational output than the rover's primary computer [4].

This was the first time a non-radiation-hardened processor was used as a computing unit on another planetary surface mission [4]. The localization pipeline was executed on the co-processor which received panoramic image data from the rover NavCams and performed visual feature extraction, estimating the pose of the rover and comparing the processed terrain data with orbital imagery to improve the localization accuracy that enabled Perseverance to determine its position with approximately 25 centimeters without requiring direct guidance from Earth [4, 5]

4 The Software Behind Mars Global Localization – How the "Censible" Framework Works

As a rover drives across the Martian surface, it accumulates errors in its position estimate (odometry uncertainty). Without a way to rectify this drift, the rover cannot be certain of its true location, which limits its ability to navigate safely and reach scientifically valuable targets. Previously, resetting this positional uncertainty required a time-consuming manual process where human operators on Earth compared the rover images to orbital maps. The Mars Global Localization (MGL) technology, built on the "Censible" Framework, automates this process directly onboard the Perseverance rover, achieving GPS-like autonomy on Mars.

4.1 The "Censible" Framework

Rover localization is modeled as an image registration problem between the rover images and the orbital map. A model-based approach is taken to make the rover images match the orbital map image as closely as possible, then find a similarity measure that is invariant to the differences. By running one of these activities in advance, the registration problem can be simplified to a search over 2D translation [7]. This is split into three main steps:

Step 1: Panoramic Imaging & Radiometric Correction:
A 360 panorama of stereo pairs around the rover is acquired to maximize the area of imaged terrain. The image formation process from pixel values to scene radiances is modeled. A key motivation for this stage is to make the lighting consistent across all images in the combined orthomosaic. The equations for this are as follows:

$$p(x, y) = \tau_{\text{comp}}(f(x, y)\, r_s(x, y)\, t_{\text{exp}} + b) \tag{1}$$

$$r_s(x, y) = \frac{\tau_{\text{comp}}^{-1}\big(p(x, y)\big) - b}{f(x, y)\, t_{\text{exp}}} \tag{2}$$

Step 2: Top-Down Orthomosaic Projection:
Images are stereo rectified, and stereo matching is performed using semi-global matching. The stereo point cloud is rotated to align with the orbital frame, using the previously recovered rover absolute orientation and the mast joint angles. These points are filtered to between 3 and 40 meters to remove the rover body and shadows in the near range, and noisy stereo in the far range. The stereo point clouds are ortho-projected into a 25cm resolution pixel grid to match the orbital appearance maps.

Step 3: Cross-Referencing with Orbital Maps:
The final step is that the orbital map is cropped around the rover's position estimate from odometry, with a width of the rover's position uncertainty plus the stereo orthomosaic width. Both the template and orbital map undergo a modified census transform, a non-parametric local image transform that is invariant to monotonic variations in intensity.

$$c(p, p') = \begin{cases} 0 & \text{if } p > p' \\ 1 & \text{if } p \leq p' \\ \emptyset & \text{if either } p \text{ or } p' \text{ have no data} \end{cases} \tag{3}$$

The proposed algorithm uses a modified Hamming distance, which does not penalize missing data, which comes from geometric occlusions and stereo failures that are a natural consequence of the rover's perspective:

$$H(d, d') = \begin{cases} 0 & \text{if } d = d',\ d = \emptyset \text{ or } d' = \emptyset \\ 1 & \text{otherwise } (d \neq d') \end{cases} \tag{4}$$

The objective of template matching is to establish which x, y locations minimize (SC) and (SE), indicating that the template and image region match.

$$\min_{x,y \in 1..240} S_C(x, y) + \alpha S_E(x, y) \tag{5}$$

where α enables for a weighted sum of the census and elevation scores, with a greater weight placed on the census scores. The minimizing (x, y) pixel coordinates are then converted to meters and then to a global northing and easting.

The accuracy of the approach above is fundamentally limited by map resolution, so we estimate the subpixel location by fitting a 2D quadratic surface model to the 3x3 pixel neighborhood surrounding the best score:

$$p(x, y) = \sum_{i=0}^{2} \sum_{j=0}^{2} a_{i,j} x^i y^j \tag{6}$$

To correctly translate map vectors to rover vectors, the east-west scaling error can be compensated by the formula:

$$\text{easting}_{\text{map}} = \frac{\text{easting}_{\text{rover}} \cdot \cos \varphi_{ts}}{\cos \varphi}$$

where φ is the latitude of the rover and φ_{ts} is the latitude of the true scale (or standard parallel). No scale correction is required for the northing component.

4.2 The Enhanced Accuracy and Autonomy

This fully automated process allows the Perseverance rover to independently determine its precise location. The Result: It pinpoints the rover to within 25 centimeters (10 inches) in just two minutes, all without human help [4]. This is a significant improvement over the previous manual method, which required communication relays between Earth and Mars. Running this software on a repurposed processor originally used for the Ingenuity Mars Helicopter, the rover can now autonomously reset its position uncertainty, allowing it to carry longer and more precise drives without human interventions.

5 Results on the Perseverance Rover

This framework was implemented on Mars on the Perseverance rover and successfully executed on Martian sol 914. The system was executed in shadow mode, using images from a previous sol without updating the pose. The binary was 744KB, with a small 250KB appearance and an elevation orbital map representing a 200m x 200m area.

Algorithm	**Avg. Error**	**99%-Tile**	**>5m Errors**
Masked NCC	24.97m	45.90m	225/264
Masked Mutual Information	24.33m	46.98m	216/264
Modified Census (Ours)	**0.36m**	**0.93m**	**0/264**

Table 1: Comparison of similarity measures, demonstrating the sub-meter accuracy of the chosen Modified Census algorithm [7].

Resolution	Pairs	Total	Stereo	Matching	Memory
2560x1920	1	36s	14s	8s	<1GB
2560x1920	5	81s	65s	8s	<1GB

Table 2: Execution timing on flight-identical hardware, showing the algorithm completes within two minutes using minimal volatile memory [7].

6 Conclusion

The successful implementation of Mars Global Localization marks an important milestone in the realm of autonomous planetary exploration. By overcoming the limitations of visual odometry drift and the severe communication latency of "ground-in-the-loop" operations, the Perseverance rover is now capable of navigating the Martian surface with independence. The re-purposing of the Ingenuity Helicopter Base Station provided the necessary computational bandwidth to process high-resolution NavCam orthomosaics and cross-reference them against orbital maps. This framework provides human level accuracy. It gives the rover the ability to drive potentially unlimited distances without requiring localization from Earth.

7 Future Works

Beyond Perseverance, such a system is key for future planetary robotic missions. This framework expands the operational capabilities for future rovers, as space agencies plan for exploration in environments with even greater communication delays or complex terrains, such as the icy moons of Europa and Titan, or the highly time-sensitive Mars Sample Return campaign, the ability for a robotic explorer to autonomously pinpoint its location will be paramount. Ultimately, the success of MGL on Perseverance proves that the future of extraterrestrial exploration relies just as heavily on intelligent, adaptable software architecture as it does on mechanical engineering.

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